Differential Inclusion Problem

Ordinary Differential Equation (ODE):

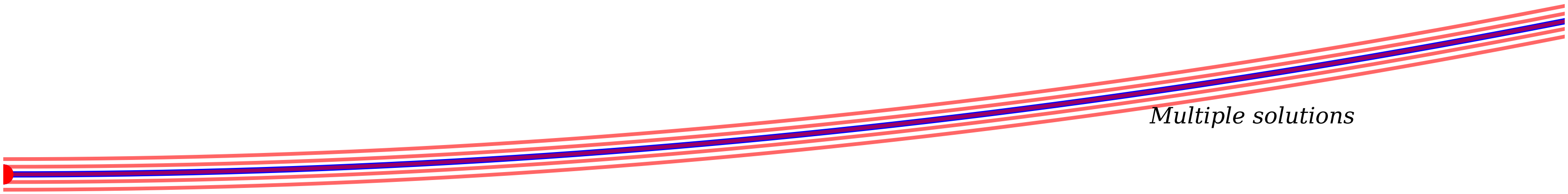
$$\frac{dx}{dt} = f(t, x(t))$$

Differential Inclusion (DI):

$$\frac{dx}{dt} \in F(t, x(t))$$

Single trajectory (ODE):

Solution set (DI):



Key Differences:

- ODE: Single unique solution (under Lipschitz conditions)
- DI: Multiple possible solutions (solution set)
- DI: Includes uncertainties, delays, switching controls
- DI: Applications: optimal control, delay systems